

ZONO — Master System Audit (Codebase Reality Map)

Audit date: 2026-06-20 **Scope:** Read-only audit of `/Users/talzatelman/Desktop/zono-os` — actual repository + Supabase migrations only. **Method:** Repo scan (routes, services, engines, migrations) + live migration replay + `tsc` / `eslint` gates. No features built, no code changed, no migrations added.

Honesty note: Everything below is grounded in files/tables/routes that exist in the repo. Items that could not be confirmed are marked **unclear** / **needs manual verification**. Anything not present in code is marked **not found in codebase**.

1. Executive Summary

ZONO is a **deterministic, Hebrew/RTL, multi-tenant real-estate intelligence SaaS** built on Next.js (App Router, `(app)` route group) + Supabase (`@supabase/ssr` , org-scoped RLS via `current_org_id()` / `has_min_role()`) + Tailwind v4 + TypeScript.

The system is substantially built and internally consistent. There are **45 migrations / 143 tables** (live replay), **~40 page routes**, **6 API/cron routes**, and **~33 library modules**. A central **Decision Brain** aggregates **~32 tables** from every intelligence module, and a **Knowledge Graph** unifies entities/relationships across the CRM. The intelligence layers are deterministic (rule/scoring engines), consistent with the product's "no LLM" principle — **with one exception** (see below).

Build health: `tsc --noEmit` clean; `eslint` 0 errors (8 cosmetic unused-var warnings); all 45 migrations replay cleanly against real Postgres.

Most important honest findings: 1. **One real LLM call exists** — `src/lib/properties/ai.ts` calls OpenAI `gpt-4o-mini` for *marketing copy only*, gated behind `OPENAI_API_KEY` , server-only, with a deterministic Hebrew template fallback. This is the only real external AI call in the codebase. It does **not** touch any scoring/intelligence engine, but it does contradict a strict reading of "no AI calls." 2. **Several modules are "data + engine + dashboard" complete but depend on manual data entry** (communication, social, distribution) because real connectors (WhatsApp/Meta/Gmail) are intentionally not built. 3. **A cluster of ~12 "future" tables** exists in the distribution/social migration with schema but little/no service logic yet. 4. **Recon table-count discrepancy resolved:** a sub-scan counted 148 `create table` statements; the authoritative live replay yields **143 public tables**. Use 143.

2. Overall Completion Estimate

These are code-grounded estimates (presence of service + engine + UI + integration), not product-vision estimates.

Bucket	Approx. share of modules	Notes
Built (service + engine + UI + integrated)	~70%	Core CRM + most intelligence brains + transactions
Partially built (works but manual-data-dependent or thin UI)	~22%	communication, distribution, social-leads, acquisition surfacing
Placeholder / future (schema present, little/no logic)	~8%	~12 distribution/social future tables, broker discovery
Missing (named in audit prompt, not in codebase)	see §10	portals, signatures, calculators, automation UI, voice, etc.

Headline: roughly **70% built / 22% partial / 8% placeholder** at the module level, with a meaningful set of audit-prompt concepts simply **not present** in code (these are not "incomplete" — they were never started).

3. Module Registry

Status legend: **built** / **partial** / **placeholder** / **missing** / **unclear**. Completion % is code-grounded.

Core CRM / Infrastructure

Module	Path	Status	%	Tables	Dec.Brain	Graph	Dashboard	Notes
Supabase/clients	src/lib/supabase	built	100	—	—	—	—	client/server/service-role/middleware
Repositories (org/user/locality/props)	src/lib/repositories	built	100	organizations, users, roles, *_operating_localities, israel_localities	—	—	—	service-role for privileged writes
Auth/session	src/lib/auth	built	100	users (via auth)	—	—	—	signUp/signIn/session context
Onboarding	src/lib/onboarding	built	90	users, organizations, roles, operating_localities	—	—	—	8-step wizard
Dashboard context	src/lib/dashboard	built	95	users/orgs/localities	—	—	host	error-safe empty context
Properties	src/lib/properties	built	90	properties, property_media	via intel	yes (node)	yes	OpenAI optional in ai.ts
Buyers	src/lib/buyers	built	95	buyers	via intel	yes	yes	
Sellers	src/lib/sellers (+ service360, propertySellers)	built	90	sellers, property_sellers	via intel	yes	yes	360 profile
Tasks	src/lib/tasks	built	95	tasks	—	—	yes	logs activity
Journey	src/lib/journey	built	90	property_journeys, properties	—	—	yes	stage engine + trigger
Localities search	src/lib/localities	built	100	israel_localities	—	yes (locality nodes)	—	
Activity layer	src/lib/activity	built	100	activity_events, entity_relationships, communication_threads/messages, meetings	writes	feeds edges	—	unified event log

Intelligence Brains

Module	Path	Status	%	Tables	Dec.Brain	Graph	Dashboard	Notes
Property Intelligence	src/lib/intelligence	built	88	property_intelligence_profiles + 8 (missions/levers/risks/exposure/touchpoints/calendar/score_events/blueprints)	yes	indirect	yes	+ transaction-valuation risks (recent)
Seller Intelligence	src/lib/seller-intelligence	built	88	seller_intelligence_profiles, seller_missions/risks/touchpoints/commitments	yes	indirect	via team	+ price-gap risk (recent)
Buyer Intelligence	src/lib/buyer-intelligence	built	90	buyer_intelligence_profiles, buyer_missions/risks/touchpoints/objections/commitments	yes	indirect	via matching	
Matching Intelligence	src/lib/matching-intelligence	built	90	match_intelligence_profiles, match_risks/objections/opportunities, revenue_signals	yes	yes (edges)	yes	
Decision Brain	src/lib/decision-intelligence	built	88	decision_intelligence_profiles, attention_items, opportunity_signals, decision_queue, decision_recommendations	hub	yes (signals)	yes	aggregates ~32 tables
Communication Intelligence	src/lib/communication	partial	85	communication_intelligence_profiles, _commitments, _followups, _insights	yes	—	yes	manual logging only (no real channels)
Knowledge Graph	src/lib/graph	built	90	graph_entities, graph_relationships, graph_signals	feeds	hub	via signals	clusters + opportunity signals
Deal Forecast	src/lib/forecast	built	92	deal_forecasts, pipeline_snapshots, deal_forecast_signals	yes	yes (signals)	yes	+ pricing-gap signal (recent)

Module	Path	Status	%	Tables	Dec.Brain	Graph	Dashboard	Notes
Deal Execution	src/lib/deals	built	90	deal_profiles, deal_journeys, deal_negotiations, deal_objections, deal_tasks	yes	yes (node)	yes	
Team Intelligence	src/lib/team	built	90	team_intelligence_profiles, team_performance_snapshots, agent_coaching_signals, office_intelligence_profiles, team_opportunity_leaks, management_actions	yes	—	yes	
Lead Routing	src/lib/routing	built	90	agent_intelligence_profiles, agent_locality/property_type_performance, lead_routing_profiles/candidates	indirect	—	own page	agent-own visibility
Revenue Intelligence	src/lib/revenue	built	92	organization_revenue_profiles, revenue_targets, revenue_leakage_events	yes	—	yes	+ overpriced-inventory leak (recent)
Market Intelligence / Heatmap	src/lib/market	built	95	market_area_snapshots	yes	yes (localities)	yes	deterministic, org-scoped
Transactions Intelligence	src/lib/transactions	built	90	property_transactions, geo_coverage_targets, transaction_sync_logs, property_research_reports, building_intelligence, street_intelligence, transaction_opportunity_radar_alerts	yes (radar)	yes (research links)	yes	Apify GovMap/Madlan; pipeline research recompute drives the recent deep-integration
Geo Intelligence	src/lib/transactions/geo.ts + israel_neighborhoods	partial	70	israel_neighborhoods	—	—	coverage page	OSM neighborhood discovery (deterministic)
Broker Intelligence	src/lib/broker	built	85	broker_profiles, broker_aliases/sources/service_areas/discovery_runs/match_reviews, property_broker_matches, broker_logo_assets	yes (reviews)	yes (node)	own page	discovery run = future
Competitor Intelligence	src/lib/competitor	built	92	competitor_profiles, competitor_market_positions, competitor_signals	yes	yes (signals)	own page	
Acquisition Intelligence	src/lib/acquisition	built	85	inventory_acquisition_profiles (+ actions/reviews)	yes	—	own page	+ transaction-valuation sub-score (recent)
External Listings	src/lib/external-listings	built	90	external_listing_sources/listings/history/duplicates, import_jobs/logs	yes	—	properties tab + market	real Apify Yad2/Madlan

Marketing / Growth

Module	Path	Status	%	Tables	Notes
Marketing Intelligence	src/lib/marketing	built	90	community_profiles, community_intelligence_profiles, property_marketing_profiles, buyer_segments, marketing_opportunity_signals	feeds Decision Brain + dashboard
Distribution Workspace	src/lib/distribution	partial	75	community_dna_profiles, property_community_matches, distribution_plans/items, daily_distribution_batches/items, distribution_opportunity_signals (+ ~12 future tables)	manual publishing only , no Meta API
Daily Assisted Distribution	/distribution/daily	partial	75	daily_distribution_*	UI present
Social Lead Capture	src/lib/social-leads (+ social_* tables)	partial	75	social_interactions, social_leads, social_followups	structured leads; capture is manual
Community Attribution / ROI / Network	distribution migration	placeholder	20	community_lead_attribution, community_deal_attribution, community_rankings, community_network_profiles, community_metrics	schema only — needs cross-check with services
Content Generation	src/lib/distribution/content.ts	built	85	—	deterministic post variants (6 angles), no LLM
Content Performance / Campaigns	—	missing	—	—	not found in codebase as a dedicated module
WhatsApp/Social Messaging	—	placeholder	10	social_accounts, social_connection_vault, distribution_queue	future scaffolding only

Module	Path	Status	%	Tables	Notes
AI Reply Suggestions	src/lib/communication (draftReply)	partial	60	—	deterministic draft builder, not an LLM

Operations / Presentation / Future (audit-prompt concepts)

Concept	Status	Evidence
Digital Signatures	partial (schema only)	documents table has signature flow fields; no signing service/UI found
Document Intelligence / Contract Intelligence / Legal Workflow	missing	not found in codebase
Mortgage / Real-estate Calculators	missing	not found in codebase
Call Summary AI / Meeting Intelligence	partial	meetings table + activity; no AI summary service
Follow-Up Automation / Workflow Automation / Automation Rules	partial (schema only)	automations table + automation_trigger / automation_status enums exist; no automation engine/UI found
Client/Seller/Buyer/Agent/Manager Portals	missing	not found in codebase (single internal app only)
Multi-Branch / Branch Manager	partial	roles include manager/admin/owner + revenue_targets scope=branch; no dedicated branch UI
Office/Agent Website, Public Property Pages, Landing Pages	missing	not found in codebase
Recommendation Intelligence/Packages/Map, Territory Expansion, Referral Intelligence, Office Growth, Autonomous Office AI, AI Orchestrator, AI Agents, Reinforcement/Learning, Voice AI	missing	not found in codebase

4. Database Inventory

- **Migrations:** 45 .sql files, timestamp-ordered, no ordering violations (every FK target exists before reference; all ALTERs additive/nullable).
- **Tables (live replay):** 143 public tables.
- **Enums:** ~43 (org_plan, region, property_type/status, journey_stage, broker_, listing_source_type, lead_, deal_*, etc.).
- **Helper functions:** role_rank, current_org_id, current_role_key, has_min_role, is_org_member, set_updated_at, journey_stage_for_status, journey_progress_for_stage, create_property_journey, seed_org_default_roles.
- **Triggers:** ~130 set_updated_at triggers; append-only tables (activities, touchpoints, logs, score_events) intentionally have none.

Table groups (creating migration noted in detail by recon): - **Core/CRM:** organizations, roles, users, buyers, sellers, projects, units, properties, leads, deals, opportunities, matching_results, tasks, notes, meetings, documents, notifications, activities. - **Reference/National (global, service-role write):** israel_localities, israel_neighborhoods; property_blueprints system rows (org_id NULL). - **Operating areas:** organization_operating_localities, user_operating_localities. - **Property intelligence (9):** property_intelligence_profiles, blueprints, missions, levers, risks, exposure_channels, seller_touchpoints, calendar_plans, score_events. - **Activity/Comm:** activity_events, entity_relationships, communication_threads/messages, communication_intelligence_profiles/commitments/followups/insights. - **Seller/Buyer/Match/Decision intelligence:** as listed in §3. - **External/Broker/Competitor/Acquisition:** external_listing_, import_job_, broker_, competitor_, inventory_acquisition_. - **Routing/Team/Forecast/Revenue/Deal:** agent_, lead_routing_, team_, office_intelligence_profiles, management_actions, team_opportunity_leaks, deal_forecasts/pipeline_snapshots/deal_forecast_signals, organization_revenue_profiles/revenue_targets/revenue_leakage_events, deal_profiles/journeys/negotiations/objections/tasks. - **Marketing/Community/Distribution/Social:** community_, property_marketing_profiles, buyer_segments, marketing_opportunity_signals, distribution_, daily_distribution_, social_. - **Transactions/Geo:** property_transactions, geo_coverage_targets, transaction_sync_logs, property_research_reports, building_intelligence, street_intelligence, transaction_opportunity_radar_alerts, israel_neighborhoods.

RLS: Org-scoped tables enforce current_org_id() + role checks for SELECT/INSERT/UPDATE/DELETE. Reference tables are read-all / service-role-write. social_connection_vault is service-role only (authenticated users have no access) — correct for a future token vault.

Possibly-orphaned / future tables (schema present, thin/no service usage — needs cross-check): social_accounts, social_connection_vault, community_discovery_runs, community_discovery_candidates, community_metrics, community_lead_attribution, community_deal_attribution, community_rankings, community_network_profiles, distribution_queue, broker_discovery_runs. (~11-12 tables.)

5. Route / UI Inventory

~40 page routes under `src/app/(app)` + `(auth)` + onboarding. All app pages set `dynamic = "force-dynamic"` and read from real services (with empty-state/error fallbacks). No page found that fabricates intelligence data.

Sidebar-linked (21): `/`, `/command`, `/market`, `/properties`, `/buyers`, `/sellers`, `/matches`, `/deals`, `/transactions`, `/transactions/streets`, `/transactions/radar`, `/forecast`, `/revenue`, `/acquisition`, `/competitors`, `/marketing`, `/distribution`, `/social-leads`, `/routing`, `/team`, `/graph`.

Exist but NOT in sidebar (detail/child/admin): `/properties/[id]` (+/edit,/new), `/buyers/[id]` (+/edit,/new), `/sellers/[id]` (+/new), `/matches/[id]`, `/team/[id]`, `/broker-intelligence` (+/[id]), `/competitors/[id]`, `/external-listings/[id]`, `/transactions/coverage`, `/transactions/debug`, `/distribution/daily`, plus `/login`, `/signup`, `/onboarding`, `/auth/logout`.

Note: `/broker-intelligence` is a full page with data but is **not in the sidebar** — reachable only via drill-down. Possible navigation gap (not a data gap).

Key routes → data source (all real): `/command` → `getExecutiveCommandCenter+getTodaysFocus`; `/forecast` → `getForecastBoard`; `/revenue` → `getRevenueBoard`; `/team` → `getTeamBoard`; `/routing` → `getRoutingBoard`; `/market` → `getCurrentMarketHeatmap`; `/transactions` → `getTransactionsBoard`; `/transactions/radar` → `getRadarBoard` (+ "נתח צגרת מול עסקאות" recompute button); `/acquisition` → `getAcquisitionBoard+CC`; `/graph` → `getGraphBoard`; `/social-leads` → `getSocialLeadsBoard`; `/distribution` → `getDistributionBoard`.

UI-only / needs verification: Home `HeroSection` map pins / assistant text may be static — **unclear / needs manual verification**. `JourneysSectionContainer` data source on home — **unclear / needs manual verification**.

API/cron routes (6): `/api/external-listings/import-yad2|import-madlan|import-all|debug-provider` (session/role gated), `/api/cron/external-listings-sync`, `/api/cron/transactions-refresh` (both `CRON_SECRET` gated).

6. Real vs Mock Inventory

Finding	Detail
<code>src/data/mock.ts</code>	Used only for navigation items (and static shell scaffolding), not for business data.
Intentional fallbacks	Transactions service has a dev-only, clearly-marked mock when <code>APIFY_TOKEN</code> is missing; boards return empty objects on load failure (safe).
Real external data	External listings (Apify Yad2/Madlan) and transactions (Apify GovMap/Madlan) are real provider integrations.
Manual-data-dependent (UI+engine real, data entry manual)	communication (no WhatsApp/Gmail connector), social-leads (manual capture), distribution (manual publishing).
Placeholder tables w/o UI	the ~12 future community/social tables (\$4).
UI without real data	none confirmed; home hero visuals unclear / needs manual verification .
Real data without UI	<code>pipeline_snapshots</code> , parts of <code>community_*</code> attribution exist in schema but limited surfacing — needs manual verification .
Production accidentally using mock	low risk — mock transaction fallback is <code>NODE_ENV !== production</code> gated; verify <code>APIFY_TOKEN</code> is set in prod.
Real LLM	<code>properties/ai.ts</code> → OpenAI <code>gpt-4o-mini</code> , only when <code>OPENAI_API_KEY</code> set, server-only, deterministic fallback. Marketing copy only.

7. Integration Matrix

Y = integrated, — = no, (i) = indirect/via another brain.

Module	Decision Brain	Graph	Home Dashboard	Entity Pages	Revenue	Forecast	Team
Property Intelligence	Y	(i)	Y	Y	—	Y(feeds)	—
Seller Intelligence	Y	(i)	(i)	Y	—	Y	—
Buyer Intelligence	Y	(i)	(i)	Y	—	Y	—
Matching	Y	Y	Y	Y	Y	Y	—
Communication	Y	—	Y	Y	—	Y(feeds)	—
Activity	writes	feeds edges	—	Y	—	—	—
Forecast	Y	Y	Y	match page	Y	—	(i)
Revenue	Y	—	Y	—	—	uses fc	(i)
Deals	Y	Y	Y	deal/match	Y(feeds)	Y(feeds)	—
Team	Y	—	Y	/team	(i)	(i)	—
Routing	(i)	—	own page	/routing	—	—	(i)
Market	Y	Y	Y	—	—	Y(feeds)	—
Transactions	Y(radar)	Y(research)	Y	property/acq	Y(feeds)	Y(feeds)	—
Acquisition	Y	—	(i)	own page	(i)	—	—
Competitor	Y	Y	own page	/competitors	—	—	—
Broker	Y(reviews)	Y	(i)	broker page	—	—	—
Marketing	Y	—	Y	—	—	—	—
Distribution	Y	—	Y	—	—	—	—
Social Leads	Y	—	Y	own page	—	—	—
External Listings	Y	—	Y(market)	properties tab	—	—	—

Decision Brain is the strongest hub (reads ~32 tables across all brains). **Activity layer** writes everywhere but is not itself read by the Decision Brain (by design). **Routing** is comparatively siloed (own page; not surfaced in Decision Brain/Graph) — possible future integration point, not a defect.

8. Security / RLS / Server Findings

Area	Status	Detail
<code>createServiceRoleClient</code> usage	safe	7 files: org/user/locality privileged writes, property archive, transactions batch upsert, market snapshot upsert, external-listings sync. All server-side/batch; none exposed to client routes.
Cron auth	safe	both <code>/api/cron/*</code> require <code>Bearer \${CRON_SECRET}</code> ; disabled (401) when secret unset.
Secret exposure	safe	<code>SUPABASE_SERVICE_ROLE_KEY</code> , <code>CRON_SECRET</code> , <code>APIFY_TOKEN</code> , <code>OPENAI_API_KEY</code> read from <code>process.env</code> server-side only. <code>NEXT_PUBLIC_*</code> limited to Supabase URL/anon key + Google Maps key (browser-safe by design).

Area	Status	Detail
LLM call	needs review (policy)	properties/ai.ts real OpenAI call — gated + fallback, but contradicts strict "no AI". Decide if acceptable.
Social token vault	safe (unused)	social_connection_vault service-role only; no real OAuth flow implemented yet.
RLS / cross-org	safe	org-scoped tables filtered by current_org_id() ; reference tables read-all. No cross-org leakage path found.
Agent vs manager visibility	safe	routing/deals respect agent-own vs manager-all (e.g., deals "managers see all, agents see own").
APIFY token	safe	server-only; service isolates/skips when missing. Action: a live Apify token was pasted in chat earlier in development — rotate/revoke it if not already done (not stored in repo).

No critical/build-breaking security issue found. Nothing changed.

9. Duplication / Overlap Findings

Pair	Verdict	Rationale
activity_events vs communication_messages	healthy	event log vs threaded message history; comm logging also emits activity events.
matching-intelligence vs deals	healthy	scored pairs (upstream) vs promoted deals in execution (downstream); deal_profiles.match_id links them.
revenue vs forecast vs deals	healthy	forecast = per-deal probability; revenue = org aggregate/gap; deals = execution. Clear layering.
market vs transactions	healthy	market = our active supply/demand snapshot; transactions = real closed-deal ground truth feeding market scoring.
marketing vs distribution	unclear / verify	both emit "opportunity_signals"; marketing is entity/audience-level, distribution is property/community-level. Confirm they don't double-surface the same opportunity.
acquisition vs external-listings	healthy	raw ingested feed vs acquisition-scoring overlay (inventory_acquisition_profiles.external_listing_id).
seller touchpoints vs communication commitments	healthy	passive contact log vs active promise/obligation tracking.
israel_localities vs israel_neighborhoods	healthy	parent cities vs child neighborhoods; no cities / neighborhoods duplicate table found.

No consolidation recommended now except verifying marketing-vs-distribution signal overlap. **Do not refactor as part of this audit.**

10. Missing Modules (named in audit prompt, not found in codebase)

Confirmed not present as built modules/routes/services: Client/Seller/Buyer/Agent/Manager **Portals**; Office/Agent **Website**; **Public Property Pages**; **Landing Pages**; **Mortgage/Real-estate Calculators**; **Document Intelligence / Contract Intelligence / Legal Workflow**; **Call Summary AI / Meeting Intelligence (AI)**; **Workflow Automation engine/UI** (only automations schema + enums exist); **Campaigns / Content Performance**; **Recommendation Intelligence/Packages/Map**; **Territory Expansion**; **Referral Intelligence**; **Office Growth Intelligence**; **Autonomous Office AI / AI Orchestrator / AI Agents / Reinforcement-Learning / Voice AI**; real **WhatsApp/Meta/Gmail messaging** connectors.

Partial/scaffolding only: Digital Signatures (schema in documents), Multi-Branch (roles + revenue scope), Social Messaging (vault/queue tables).

11. Recommended Roadmap (based only on code reality)

- Decide the OpenAI policy.** Either (a) formally allow the gated marketing-copy LLM and document it, or (b) remove the OpenAI branch and keep the deterministic template. Today it's an undocumented exception to "no LLM."

2. **Finish the deterministic chain you already started** (transactions deep-integration): expose a single org-wide "recompute everything" path and ensure the radar/acquisition/forecast recompute runs after each transaction sync (currently manual via the radar button).
3. **Surface** `/broker-intelligence` (and `/transactions/coverage`) **in navigation** if they're meant to be first-class — they exist but are sidebar-hidden.
4. **Verify marketing vs distribution signal overlap** and the home `Journeys` / `Hero` data sources (the only "unclear" UI items).
5. **Convert the ~12 future community/social tables into either (a) real logic or (b) clearly-labeled "coming soon"** so they're not mistaken for built features.
6. **Only then** consider net-new modules (portals, calculators, automation engine) — these are greenfield, not completions.

12. Immediate Risks

1. **Policy/compliance:** the OpenAI marketing call is the single deviation from "no AI" — clarify before launch.
2. **Operational:** ensure `APIFY_TOKEN` and `CRON_SECRET` are set in production; without `APIFY_TOKEN`, transactions/external sync no-op (and dev mock must never reach prod — it's gated, verify).
3. **Secret hygiene:** rotate the Apify token that was shared in chat during development if not already done.
4. **Manual-data modules can look "empty"** in a fresh org (communication, social, distribution) — expected, but set user expectations.
5. **Navigation gaps** (broker-intelligence, coverage) could make built features appear missing.

13. What NOT to Build Yet

- Do **not** start portals, public websites, calculators, contract/legal AI, voice, or automation engine until the **partial** modules (communication channels, distribution publishing, social capture, geo neighborhood coverage) are either finished or explicitly deferred — they already have schema + UI waiting for data.
- Do **not** add new "opportunity_signals" producers until marketing-vs-distribution overlap is confirmed.
- Do **not** add more `community_*` future tables; wire up the ~12 that already exist or label them.

14. Exact Next Recommended Step

Make a single explicit decision on the OpenAI marketing call in `src/lib/properties/ai.ts` (allow + document, or remove), because it is the only thing in the codebase that contradicts the product's core "deterministic / no-LLM" claim. Everything else is internally consistent and can proceed on the deterministic roadmap above. This is a 1-file, low-risk decision and unblocks an honest "no-AI" or "AI-assisted-copy-only" positioning.

Build/Quality Gate Results (this audit)

- `npx tsc --noEmit`: **clean** (0 errors)
- `npx eslint src/lib, src/app`: **0 errors** (8 cosmetic unused-var warnings, pre-existing)
- Migration replay: **45/45 applied cleanly, 143 public tables**, all intelligence-stack tables present
- No code, schema, or features changed during this audit.